

Xinrui Wu

✉ xinruiwu.wx@gmail.com GS Google Scholar 🏠 Academic 🌐 GitHub

Education

University of Electronic Science and Technology of China (UESTC)
B.S. in Software Engineering (Embedded System) (GPA 3.63/4)

Sep 2023 - Present
Si Chuan, China

Research Experience

UESTC, CFM (*Center for Future Media*)

Jun 2024 – Present

Research Intern, Advisor: [Prof. Yang Yang](#) 

 UESTC, Sichuan

- Research focus: advancing Open-Vocabulary Object Detection (OVD) and Segmentation (OVS).
- Pioneered innovative methods to enhance object detection capabilities, and reduce false positives in OVD.
- Collaborated on developing a robust segmentation framework, improving accuracy across various datasets.
- Achieved significant progress in recognizing and segmenting objects with diverse appearances in images.

Megvii Technology Limited

Feb 2025 – Jul 2025

CV Algorithm Intern, Mentor: Dr. Wei Ge

 MEGVII, Beijing

- Proposed unified prompt representation method, transforming multi-modal inputs into visual masks to simplify alignment and improve training efficiency.
- Built large-scale automated data pipeline capable of processing billions of samples to support model training.
- Led development of open-vocabulary detection model achieving performance comparable to closed-source SOTA models.
- Successfully deployed the developed OVD model in business applications, validating its practical utility.

HKUST, VGD Group

Jul 2025 – Aug 2025

Research Intern, Advisor: [Prof. Sai-Kit Yeung](#) 

 HKUST, Hongkong

- Research focus: lightweight image segmentation and salient individual identification for marine environments.
- Proposed the MaskGuide mask-guided distillation framework, which enhances critical information and suppresses noise through a feature separation and purification process.
- The resulting Tiny-MSAM model achieves 99.4% of the Segment Anything Model's accuracy with only 0.5% of its parameters and an inference speed of 164.7 FPS.
- Accepted by IEEE Robotics and Automation Letters as a co-first-author paper.

Tsinghua, AIR | Dean: [Prof. Ya-qin Zhang](#) 

Jan 2025 – Present

Research Intern, Advisor: [Prof. Yuanchun Li](#) , [Prof. Yunxin Liu](#) 

 THU, Beijing

- Research focus: generation of controllable length Chain-of-Thought (CoT) for large language models.
- Proposed the BudgetThinker framework, centered on an innovative proportional control token mechanism.
- Via a two-stage training pipeline integrating specialized data and reinforcement learning, our model achieved SOTA on benchmarks like MATH with over 95% budget adherence success, and demonstrated generalization in embodied spatial reasoning.
- Accepted by EMNLP 2026 as a co-first-author paper.

MBZUAI | Dean: [Prof. Eric Xing](#) 

Jan 2026 – Present

Visiting Student, Advisor: [Prof. Yutong Xie](#) 

 MBZUAI, Abu Dhabi

- Research focus: optimization of clinical reasoning.
- Proposed a framework that integrates complex multi-stage clinical information to enhance structured medical reasoning.
- Explored frontier training paradigms for clinical world models.

Publication

BudgetThinker: Empowering Budget-aware LLM Reasoning with Control Tokens [↗](#)

Hao Wen^{*}, **Xinrui Wu^{*}**, Yi Sun, Feifei Zhang, Liye Chen, Jie Wang, Yifan Su, Yunxin Liu, Ya-Qin Zhang, Yuanchun Li
Accepted by **EMNLP 2026** 2026

MaskGuide: Efficient Distillation for Deployable Lightweight Segmentation in Marine Environments [↗](#)

Xinrui Wu^{*}, Ziqiang Zheng^{*}, Yiwei Chen, Zeyu Ma⁺, Yang Yang, Sai-Kit Yeung
Accepted by **IEEE Robotics and Automation Letters** 2026

Can Image Models Imagine Time? ImageTime: A Novel Benchmark for Probing Visual World Modeling Through Spatiotemporal Consistency [↗](#)

Xinrui Wu, Lichen Huang
Submitted to **WACV 2027** 2026

HiReT: Hierarchical Relational Trajectory Modeling for Longitudinal Clinical Prognosis

Xinrui Wu, Wanqi Yang, Min Xu, Yutong Xie
Submitted to **ICLR 2027** 2026

Predicting the Future Is Not Enough: Toward Action-Aware Clinical World Models

Xinrui Wu, Wanqi Yang, Min Xu, Yutong Xie
Submitted to **ICLR 2027** 2026

OCG: Optical Character Grounded Visual Reasoning with Flow Matching [↗](#)

Lichen Huang, **Xinrui Wu**
Submitted to **NeurIPS 2026 Workshop BeNTo** 2026

Scholarship

UESTC Model Student Scholarship | Top 5% *Sci-Tech Innovation* Oct 2025

UESTC Model Student Scholarship | Top 5% *Sci-Tech Innovation* Oct 2024

Prize

National Second Prize Aug 2024
C4 National College Student Network Technology Challenge

National Third Prize Jun 2025
China College Student Computer Design Competition

National Third Prize May 2025
National College Student Innovation and Entrepreneurship Competition

National Third Prize Aug 2024
C4 National College Student Network Technology Challenge

Second Prize of SiChuan Province Jun 2024
China College Student Computer Design Competition

Interests

Clinical AI: Clinical reasoning, longitudinal prognosis, relational trajectory modeling

World Models: Visual world modeling, spatiotemporal consistency, causal reasoning

Efficient Reasoning & Perception: Budget-aware LLM reasoning, lightweight segmentation, distillation

Spatial Intelligence: 3D perception, vision-language spatial reasoning, AR/VR

Generative Models: Controllable image generation, reference-guided editing, generative model evaluation